

Exam Date & Time: 09-May-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. EXAMINATIONS - APRIL / MAY 2024
SUBJECT: BIO 1071-CHM/BIO 1051-B/BIO_1071-B - BIOLOGY FOR ENGINEERS
BIOLOGY FOR ENGINEERS [BIO 1071-CHM]

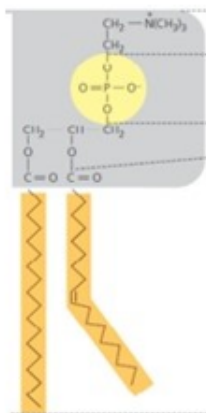
Marks: 50

Duration: 180 mins.

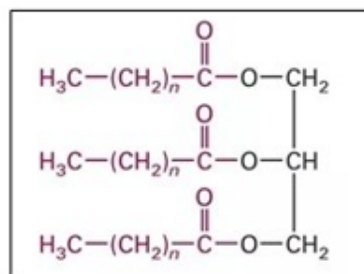
A

Answer all the questions.

- 1A) Students are given a task of designing an artificial cell membrane from the following two molecules. Explain logically the questions related to designing of the cell membrane.



Molecule A



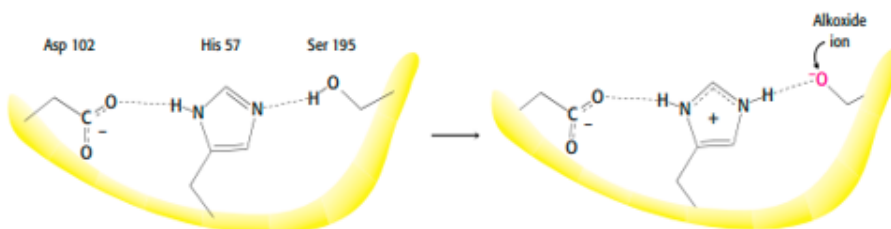
Molecule B

(4)

- Which among the following molecules given is most likely to assemble and form a lipid bilayer?
 - What property of the molecule that is selected allows it to form a lipid bilayer?
 - Which bond(s) or interaction(s) is/are most likely to stabilize the lipid bilayer?
 - Illustrate the lipid bilayer structure in the aqueous environment by the selected molecule.
- (1+1+1+1 = 4 marks)

- 1B) Justify the reason for not having a non-carbon life on earth. Replacement of Zinc by Cadmium is seen in marine algae during nutrient limiting conditions. Identify the reason for the substitution to occur. Studies conducted with Copper does not yield the same result. Justify. (3)

- 1C) Histidine, Serine and Aspartic acid form the catalytic triad of the enzyme given in the figure.

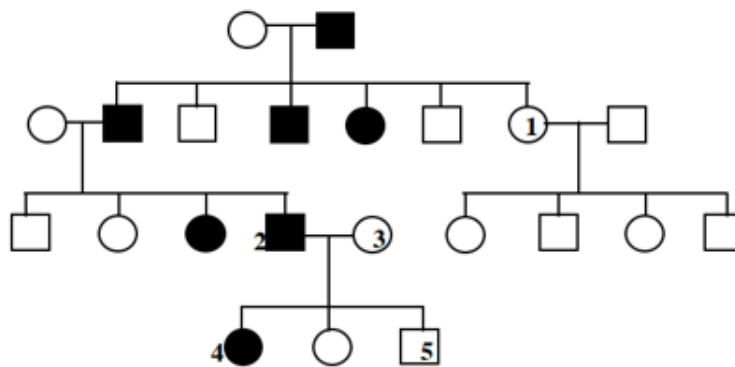


(3)

- What is the effect if serine is substituted by alanine amino acid? What happens to the catalytic power? Justify.
- If aspartic acid is substituted by threonine, what is the effect?

- 2A) i) Tomato plants can be tall or short and have notched or smooth leaves. You cross a tall, smooth leafed plant with a short, notched leafed plant. All of the progeny are tall, and notched leafed. Identify the dominant and recessive traits?
- ii) Mention the genotypes of the two true-breeding parents? Use the letters H or h to represent the alleles of the height gene and the letters S or s to represent the alleles of the leaf gene.
- iii) Two tall, notched F₁ plants were crossed (self cross) to get 1600 F₂ plants. What ratio of phenotype that you expect in the F₂ generation? If you get 400 plants in F₂, how many of these will be homozygous recessive for both traits? (4)
- iv) You cross an F₁ Plant with a double homozygous recessive plant and get 1600 plants in F₂ generation. Give the types of the F₂ plants that you will see. Give the genotype, phenotype and the number of each of F₂ classes generated.
(1+1+1+1 = 4 marks)

- 2B) Carefully analyse the following pedigree, filled symbols represent the affected individuals.



- i) What is the most likely mode of inheritance for this pedigree?
- ii) State the genotypes of individuals # 1-5 in the following table using the letter "A". Use the uppercase letter to represent the dominant allele and lowercase letter to represent the recessive allele.
- iii) If individuals #2 and #3 have another son what are the chances that this son will be affected? (3)
- 2C) In cattle, hornless (P) is dominant over horned (p). A hornless bull is mated to three cows, A, B, C. With cow A, which is horned, a hornless calf is produced. With cow B, also horned, a horned calf is produced. With cow C, which is hornless, a horned calf is produced. What are the genotypes of the four parents (Cows A, B, C and Bull). Justify (3)

- 3A) We have given the partial DNA sequence from a human. Based on the information provided, answer the following.

Base triplet number	1	2	3	4	5	6	7
Human DNA sequence	ATG	TGT	CCA	TTA	ACG	TGC	ACA

- i) Name the codon that is formed from base triplet number 2 on the DNA template sequence.
- ii) Write down the names of the amino acids coded for by base triplets 6 and 7.
- iii) If a mutation changes base triplet 1 from ATG to ATA, will it change the protein formed?
(1+2+1 = 4 marks) (4)
- 3B) Based on the sequence, answer the following: (3)

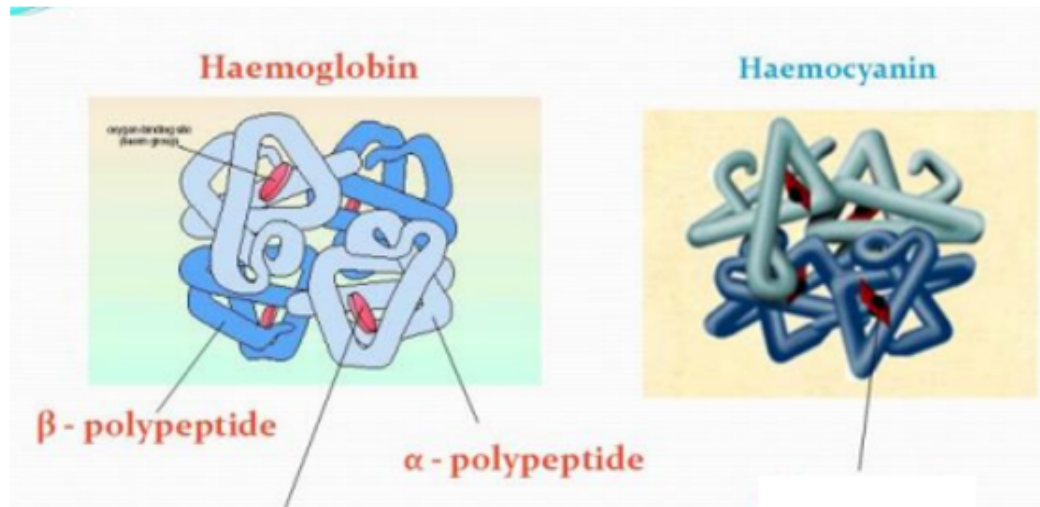
5' ACUAGCAGGAGACGUAAGCGAUGUGCCAGAUGCGCAGUCACACAUAACUGCAAG 3'

- i) How many amino acids long is the protein?
 - ii) How many tRNAs will be required to make this protein
 - iii) Mention the second amino acid in the protein (N terminus to C terminus)
- (1+1+1 = 3 marks)

3C) Consider the following DNA sequence:
 5'ATGCCAATACCATAGCTAAATTTGGCACTCGATAA3'
 This sequence translates to the following protein sequence:
 N-Met-Pro-Ala-Lys-Phe-Gly-Thr-Arg-C (3)
 Determine the sequences of primary and mature RNA. What will happen if the underlined nucleotide mutates to "T"?

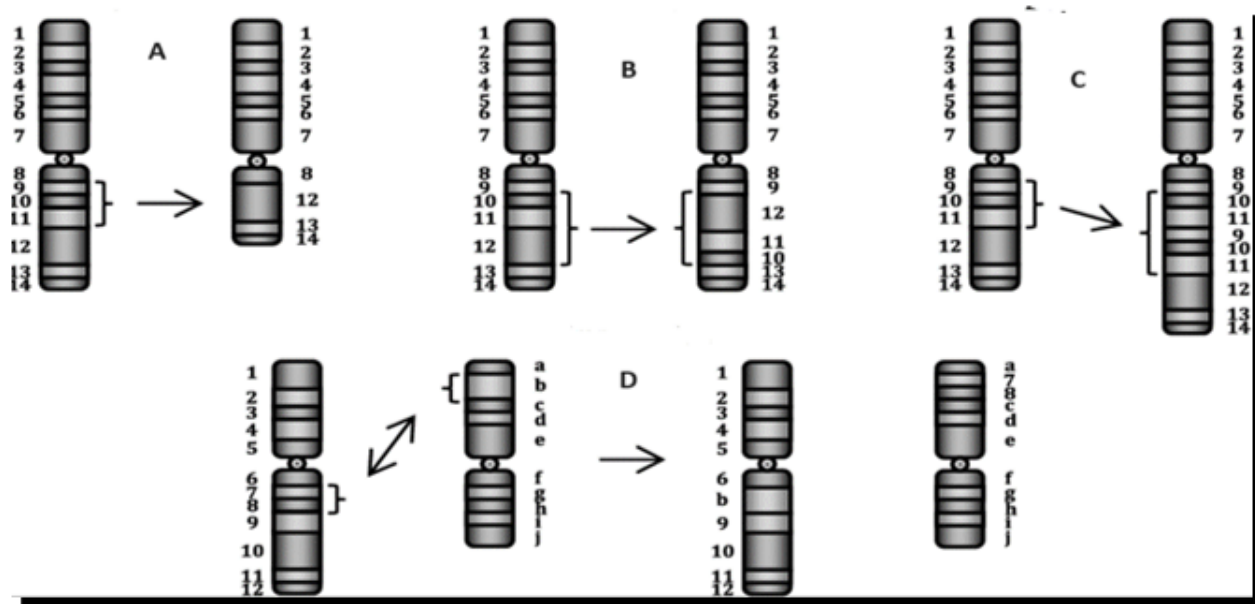
4A) Sickle cell disease (SCD) describes a group of blood conditions that affect hemoglobin in RBCs. Hemoglobin helps RBCs deliver oxygen throughout the body. SCD causes changes in the hemoglobin molecule that can affect the shape and function of RBCs. While this condition can negatively affect RBCs and cause potentially life-changing symptoms, certain types of SCD may offer protection against malaria. (3)
 i) What is the link between sickle cell anemia and malaria?
 ii) How Does Sickle Cell Protect Against Malaria?
 iii) Why does the Sickle Cell Trait Remain in the Human Population?
 (1+1+1 = 3 marks)

4B) Referring to the figure answer the following questions.



- i) How do these two molecules explain the genetic evidence for evolution?
 - ii) Elaborate on the structural difference between the two molecules given below.
 - iii) Explain why crabs, lobsters, snails, and octopuses have blue blood.
- (1+1+1 = 3 marks)

4C) Explain and identify the type of chromosomal mutation given in the figure A, B, C, and D (4)



- 5A) i) Does live attenuated vaccine requires adjuvants? Justify.
 ii) Inactivated and Live attenuated are two different types of vaccines. Which among them has less risk? Give an example of an Inactivated vaccine. (3)
 (1+2 = 3 marks)
- 5B) i) Most Vaccines require boosters. What is a booster dose? and why do most vaccines require booster doses? (3)
 ii) How do vaccines work? Are all vaccines 100% effective? Justify. (3)
 (1+2 = 3 marks)
- 5C) What is the concept for bioinspiration in the following cases? Explain. (4)

Sl No	Concept	Explanation
1		
2		
3		
4		

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